Image Processing Winter 2026-2027

Homework 1 - Histograms & Point Operations

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Questions 1–2: (80 Points)

Programming assignment — See attached notebook in Moodle.

Question 3: (12 Points)

In this question we consider the following point operation $M : [0, 1] \to [0, 1]$:

$$M(v) = \begin{cases} 0, & v \le k_1 \\ \dfrac{v - k_1}{k_2 - k_1}, & k_1 < v < k_2 \\ 1, & v \ge k_2 \end{cases} \qquad 0 \le k_1 < k_2 \le 1. \tag{1}$$

Let f be a grayscale image with values in $[0, 1]$, and let $g(x) = M(f(x))$ be the image obtained by applying M pixelwise.

1. **Invariance:** Does there exist an image f with at least three distinct gray levels that is invariant under M (i.e., $M(f) = f$ pointwise)? If yes, give an example and prove it. If not, prove that no such image exists.

2. **Mean intensity:** Let $\mu(f)$ denote the mean gray level of f. How does $\mu(g)$ compare to $\mu(f)$? Is it always larger, always smaller, or dependent on f (and on k_1, k_2)? Explain.

3. **Contrast & iteration:** Assume for the sake of this section that f contains values only in $[k_1, k_2]$, such that $g \neq f$.

   (a) Compare $\mathrm{var}(g)$ and $\mathrm{var}(f)$ (contrast). Is $\mathrm{var}(g)$ always larger, always smaller, or dependent on f? Justify.

   (b) Consider repeated application $f^{(t+1)} = M(f^{(t)})$ with $f^{(0)} = f$. Describe the limiting image as $t \to \infty$.

4. **Negative after application:** Does there exist an image h such that $M(h) = 1 - h$? If yes—construct and prove. If not—prove impossibility.

Question 4: (8 Points)

Let I_1 be a grayscale image. First apply histogram equalization to obtain I_2. Then apply a point operation on I_2 to obtain I_3.

1. Prove that the overall mapping from I_1 to I_3 is itself a point operation. That is, there exists a function M on gray levels such that $I_3(x) = M(I_1(x))$ for all pixels x.

2. Now assume the second operation, denoted Q (the one applied on I_2 to obtain I_3) is the uniform quantizer:

$$Q(g) = \left\lfloor \frac{g}{64} \right\rfloor \cdot 64 \quad \text{(bin size 64)}. \tag{2}$$

Define the overall point operation (the sequence of the two operations applied on I_1) as M.

Let I_1, J_1 be the images shown below, and let $J_3 = M(J_1)$. Given the histograms of I_1 and J_1, describe qualitatively how J_3 will look without the use of numeric computation. Explain your reasoning.

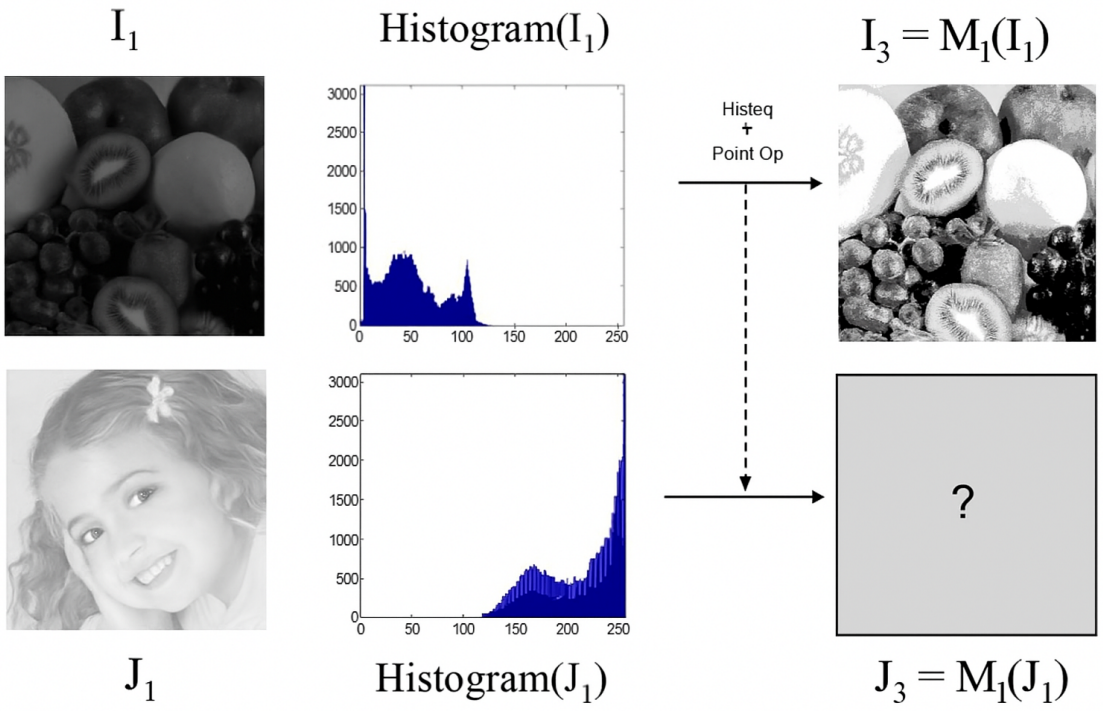


Figure 1: Illustration for Question 4. Top row: the image I_1, its histogram, and the result $I_3 = M(I_1)$ after applying Histogram Equalization followed by the point operation. Bottom row: the image J_1, and ita histogram.

Good Luck!